LaTeX Symbols for Statistics and Math

A compact personal lookup sheet

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This note is a compact lookup sheet for symbols I commonly need in mathematics, probability, statistics, and asymptotic derivations. It is inspired by the Rice LaTeX mathematical symbols handout at https://www.cmor-faculty.rice.edu/~heinken/latex/symbols.pdf, with extra emphasis on statistical notation such as stochastic convergence, orders in probability, likelihood notation, and comparison symbols such as $\lesssim$ and $\gtrsim$.

Recommended Preamble

Most entries below work with the standard AMS packages:

```
\usepackage{amsmath,amssymb,mathtools,bm}
\usepackage{mathrsfs} % for \mathscr
\DeclareMathOperator{\Var}{Var}
\DeclareMathOperator{\Cov}{Cov}
\DeclareMathOperator{\diag}{diag}
\DeclareMathOperator{\tr}{tr}
\DeclareMathOperator*{\argmax}{arg\,max}
\DeclareMathOperator*{\argmin}{arg\,min}
\newcommand{\indep}{\mathrel{\perp\!\!\!\perp}}
\newcommand{\dd}{\,\mathrm{d}}
```

Other Styles (Math Mode Only)

These commands are useful when the visual style carries mathematical meaning. Math alphabet commands usually affect only letters, digits, and uppercase Greek; use `\boldsymbol` or `\bm` when you need bold Greek letters or bold symbols.

LaTeX input	Output	Common use / note
`\mathnormal {ABCxyz012}`	$\mathnormal{ABCxyz012}$	Default math alphabet; variables and ordinary symbolic letters.
`\mathrm {ABCxyz012}`	$\mathrm{ABCxyz012}$	Upright Roman; constants, units, differential d, and short labels.
`\mathit {ABCxyz012}`	$\mathit{ABCxyz012}$	Text italic in math mode; useful for multi-letter italic labels.
`\mathbf {ABCxyz012}`	$\mathbf{ABCxyz012}$	Upright bold Latin letters and digits; common for vectors and matrices.
`\boldsymbol {\alpha +\Sigma }`	$\boldsymbol{\alpha +\Sigma}$	Bold Greek letters and math symbols; provided by AMS packages.
`\bm {\alpha +\Sigma }`	$\bm{\alpha +\Sigma}$	Robust bold math symbols; provided by the `bm` package.
`\mathsf {ABCxyz012}`	$\mathsf{ABCxyz012}$	Sans-serif math alphabet; sometimes used for random matrices or categories.
`\mathtt {ABCxyz012}`	$\mathtt{ABCxyz012}$	Typewriter math alphabet; useful for code-like labels.
`\mathcal {ABC}`	$\mathcal{ABC}$	Calligraphic uppercase letters; common for sigma fields, classes, and spaces.
`\mathbb {RNQZC}`	$\mathbb{RNQZC}$	Blackboard bold uppercase letters; number systems and probability $\mathbb{P}$.

LaTeX input	Output	Common use / note
`\mathfrak {gslABC}`	$\mathfrak{gslABC}$	Fraktur/Gothic letters; algebras, ideals, and abstract structures.
`\mathscr {FGL}`	$\mathscr{FGL}$	Script uppercase letters; often used for filtrations and function classes.
`\operatorname {logit}(p)`	$\operatorname{logit}(p)$	Upright custom operator with correct operator spacing.
`\text {if } x>0`	$\text{if } x>0$	Text inside formulas; provided by `amsmath`.
`\displaystyle \sum _{i=1}^n x_i`	$\displaystyle \sum _{i=1}^n x_i$	Display style; larger operators and limits above/below.
`\textstyle \sum _{i=1}^n x_i`	$\textstyle \sum _{i=1}^n x_i$	Inline/text style; compact operators with side limits.
`\scriptstyle x_i`	$\scriptstyle x_i$	First-level subscript/superscript size.
`\scriptscriptstyle x_i`	$\scriptscriptstyle x_i$	Second-level nested subscript/superscript size.

Greek Letters and Variants

Symbol	LaTeX input	Symbol	LaTeX input
α	`\alpha`	β	`\beta`
γ	`\gamma`	δ	`\delta`
ϵ	`\epsilon`	ε	`\varepsilon`
ζ	`\zeta`	η	`\eta`
θ	`\theta`	ϑ	`\vartheta`
ι	`\iota`	κ	`\kappa`
λ	`\lambda`	μ	`\mu`
ν	`\nu`	ξ	`\xi`
o	`o`	π	`\pi`
ϖ	`\varpi`	ρ	`\rho`
ϱ	`\varrho`	σ	`\sigma`
ς	`\varsigma`	τ	`\tau`
υ	`\upsilon`	ϕ	`\phi`
φ	`\varphi`	χ	`\chi`
ψ	`\psi`	ω	`\omega`
Γ	`\Gamma`	Δ	`\Delta`
Θ	`\Theta`	Λ	`\Lambda`
Ξ	`\Xi`	Π	`\Pi`
Σ	`\Sigma`	Υ	`\Upsilon`
Φ	`\Phi`	Ψ	`\Psi`
Ω	`\Omega`	$\digamma$	`\digamma`

Relations, Comparisons, and Asymptotic Order

Symbol	LaTeX input	Symbol	LaTeX input
$\le$	`\le`	$\ge$	`\ge`
$\leq$	`\leq`	$\geq$	`\geq`
$\leqslant$	`\leqslant`	$\geqslant$	`\geqslant`
$\ll$	`\ll`	$\gg$	`\gg`
$\lll$	`\lll`	$\ggg$	`\ggg`

Symbol	LaTeX input	Symbol	LaTeX input
$\lesssim$	`\lesssim`	$\gtrsim$	`\gtrsim`
$\lessapprox$	`\lessapprox`	$\gtrapprox$	`\gtrapprox`
$\preceq$	`\preceq`	$\succeq$	`\succeq`
$\precsim$	`\precsim`	$\succsim$	`\succsim`
$\prec$	`\prec`	$\succ$	`\succ`
$\sim$	`\sim`	$\nsim$	`\nsim`
$\simeq$	`\simeq`	$\backsimeq$	`\backsimeq`
$\approx$	`\approx`	$\thickapprox$	`\thickapprox`
$\asymp$	`\asymp`	$\eqsim$	`\eqsim`
$\equiv$	`\equiv`	$\cong$	`\cong`
$\propto$	`\propto`	$\doteq$	`\doteq`
$\coloneqq$	`\coloneqq`	$\eqqcolon$	`\eqqcolon`
$\triangleq$	`\triangleq`	$\overset{\Delta}{=}$	`\overset {\Delta }{=}`
$a_n = O(b_n)$	`a_n = O(b_n)`	$a_n = o(b_n)$	`a_n = o(b_n)`
$X_n = O_p(a_n)$	`X_n = O_p(a_n)`	$X_n = o_p(a_n)$	`X_n = o_p(a_n)`
$a_n \asymp b_n$	`a_n \asymp b_n`	$a_n \lesssim b_n$	`a_n \lesssim b_n`

Negated Relations

Symbol	LaTeX input	Symbol	LaTeX input
$\neq$	`\neq`	$\not\equiv$	`\not \equiv`
$\nless$	`\nless`	$\ngtr$	`\ngtr`
$\nleq$	`\nleq`	$\ngeq$	`\ngeq`
$\nleqslant$	`\nleqslant`	$\ngeqslant$	`\ngeqslant`
$\npreceq$	`\npreceq`	$\nsucceq$	`\nsucceq`
$\nprec$	`\nprec`	$\nsucc$	`\nsucc`
$\not\sim$	`\not \sim`	$\not\approx$	`\not \approx`
$\notin$	`\notin`	$\nsubseteq$	`\nsubseteq`
$\nsupseteq$	`\nsupseteq`	$\nmid$	`\nmid`

Set Theory and Logic

Symbol	LaTeX input	Symbol	LaTeX input
$\in$	`\in`	$\ni$	`\ni`
$\subset$	`\subset`	$\supset$	`\supset`
$\subseteq$	`\subseteq`	$\supseteq$	`\supseteq`
$\subsetneq$	`\subsetneq`	$\supsetneq$	`\supsetneq`
$\varsubsetneq$	`\varsubsetneq`	$\varsupsetneq$	`\varsupsetneq`
$\cup$	`\cup`	$\cap$	`\cap`
$\bigcup_{i=1}^n A_i$	`\bigcup _{i=1}^n A_i`	$\bigcap_{i=1}^n A_i$	`\bigcap _{i=1}^n A_i`
$\setminus$	`\setminus`	$\complement$	`\complement`
$\emptyset$	`\emptyset`	$\varnothing$	`\varnothing`
$\mathbb {N}$	`\mathbb {N}`	$\mathbb {Z}$	`\mathbb {Z}`
$\mathbb {Q}$	`\mathbb {Q}`	$\mathbb {R}$	`\mathbb {R}`
$\mathbb {C}$	`\mathbb {C}`	$\mathcal {A}$	`\mathcal {A}`
$\forall$	`\forall`	$\exists$	`\exists`
$\nexists$	`\nexists`	$\neg$	`\neg`

Symbol	LaTeX input	Symbol	LaTeX input
$\land$	`\land`	$\lor$	`\lor`
$\implies$	`\implies`	$\iff$	`\iff`
$\therefore$	`\therefore`	$\because$	`\because`

Probability and Statistics

Symbol	LaTeX input	Symbol	LaTeX input
$\mathbb {P}(A)$	`\mathbb {P}(A)`	$\Pr (A)$	`\Pr (A)`
$\mathbb {E}[X]$	`\mathbb {E}[X]`	$\Var (X)$	`\Var (X)`
$\Cov (X,Y)$	`\Cov (X,Y)`	$\Corr (X,Y)$	`\Corr (X,Y)`
$\mathbf {1}_{\{X>0\}}$	`\mathbf {1}_{\{X>0\}}`	$X \indep Y$	`X \indep Y`
$X \sim F$	`X \sim F`	$X \sim \mathcal {N}(\mu ,\sigma ^2)$	`X \sim \mathcal {N}(\mu ,\sigma ^2)`
$X \sim \mathrm {Bernoulli}(p)$	`X \sim \mathrm {Bernoulli}(p)`	$X \sim \mathrm {Binomial}(n,p)$	`X \sim \mathrm {Binomial}(n,p)`
$X \sim \mathrm {Poisson}(\lambda)$	`X \sim \mathrm {Poisson}(\lambda )`	$X \sim \chi ^2_k$	`X \sim \chi ^2_k`
$X_n \xrightarrow {p} X$	`X_n \xrightarrow {p} X`	$X_n \xrightarrow {d} X$	`X_n \xrightarrow {d} X`
$X_n \xrightarrow {a.s.} X$	`X_n \xrightarrow {a.s.} X`	$X_n \xrightarrow {L^2} X$	`X_n \overset {L^2}{\longrightarrow } X`
$X_n \rightsquigarrow X$	`X_n \rightsquigarrow X`	$X_n \overset {d}{=} Y_n$	`X_n \overset {d}{=} Y_n`
$\hat {\theta }$	`\hat {\theta }`	$\widehat {\theta }$	`\widehat {\theta }`
$\tilde {\theta }$	`\tilde {\theta }`	$\bar {X}_n$	`\bar {X}_n`
$\ell (\theta)$	`\ell (\theta )`	$\dot {\ell }(\theta)$	`\dot {\ell }(\theta )`
$\ddot {\ell }(\theta)$	`\ddot {\ell }(\theta )`	$\mathcal {I}(\theta)$	`\mathcal {I}(\theta )`
$\hat {\theta }_{\mathrm {MLE}}$	`\hat {\theta }_{\mathrm {MLE}}`	$\mathrm {se}(\hat {\theta })$	`\se (\hat {\theta })`
$\arg \max _{\theta \in \Theta } \ell (\theta)$	`\argmax _{\theta \in \Theta }\ell (\theta )`	$\arg \min _{\theta \in \Theta } L(\theta)$	`\argmin _{\theta \in \Theta } L(\theta )`

Linear Algebra

Symbol	LaTeX input	Symbol	LaTeX input
$\mathbf {x}$	`\mathbf {x}`	$\bm {\beta }$	`\bm {\beta }`
$\boldsymbol {\Sigma }$	`\boldsymbol {\Sigma }`	$\mathsf {X}$	`\mathsf {X}`
$x^\top y$	`x^\top y`	A^{-1}	`A^{-1}`
$A^\dagger$	`A^\dagger`	$A^{1/2}$	`A^{1/2}`
$\lVert x\rVert$	`\lVert x\rVert`	$\lVert x\rVert _2$	`\lVert x\rVert _2`
$\langle x,y\rangle$	`\langle x,y\rangle`	$x \otimes y$	`x \otimes y`
$x \oplus y$	`x \oplus y`	$A \odot B$	`A \odot B`
$\det (A)$	`\det (A)`	$\tr (A)$	`\tr (A)`
$\rank (A)$	`\rank (A)`	$\diag (a_1,\ldots ,a_n)$	`\diag (a_1,\ldots ,a_n)`
$\lambda _{\max }(A)$	`\lambda _{\max }(A)`	$\lambda _{\min }(A)$	`\lambda _{\min }(A)`
$A \succeq 0$	`A \succeq 0`	$A \succ 0$	`A \succ 0`
$\begin {bmatrix}a&b\\c&d\end {bmatrix}$	`\begin {bmatrix}a&b\\c&d\end {bmatrix}`	$\begin {pmatrix}a\\b\end {pmatrix}$	`\begin {pmatrix}a\\b\end {pmatrix}`

Calculus, Sums, and Integrals

Symbol	LaTeX input	Symbol	LaTeX input
∂	`\partial`	∇	`\nabla`
$\nabla_\theta f(\theta)$	`\nabla _\theta f(\theta )`	$\nabla^2_\theta f(\theta)$	`\nabla ^2_\theta f(\theta )`
$\frac{\partial f}{\partial x}$	`\frac {\partial f}{\partial x}`	$\frac{\partial^2 f}{\partial x\,\partial y}$	`\frac {\partial ^2 f}{\partial x\,\partial y}`
$\dot{x}$	`\dot {x}`	$\ddot{x}$	`\ddot {x}`
$\sum_{i=1}^n x_i$	`\sum _{i=1}^n x_i`	$\prod_{i=1}^n x_i$	`\prod _{i=1}^n x_i`
$\int_a^b f(x)\,\mathrm{d}x$	`\int _a^b f(x)\dd x`	$\iint_D f(x,y)\,\mathrm{d}x\,\mathrm{d}y$	`\iint _D f(x,y)\dd x\dd y`
$\lim_{n\to\infty} a_n$	`\lim _{n\to \infty } a_n`	$\limsup_{n\to\infty} a_n$	`\limsup _{n\to \infty } a_n`
$\liminf_{n\to\infty} a_n$	`\liminf _{n\to \infty } a_n`	$\sup_{x\in A} f(x)$	`\sup _{x\in A} f(x)`
$\inf_{x\in A} f(x)$	`\inf _{x\in A} f(x)`	$\max_{x\in A} f(x)$	`\max _{x\in A} f(x)`
$\min_{x\in A} f(x)$	`\min _{x\in A} f(x)`	$\operatorname{ess\,sup} X$	`\operatorname *{ess\,sup} X`

Arrows and Maps

Symbol	LaTeX input	Symbol	LaTeX input
$\leftarrow$	`\leftarrow`	$\rightarrow$	`\rightarrow`
$\longleftarrow$	`\longleftarrow`	$\longrightarrow$	`\longrightarrow`
$\Leftarrow$	`\Leftarrow`	$\Rightarrow$	`\Rightarrow`
$\Longleftarrow$	`\Longleftarrow`	$\Longrightarrow$	`\Longrightarrow`
$\leftrightarrow$	`\leftrightarrow`	$\Leftrightarrow$	`\Leftrightarrow`
$\longleftrightarrow$	`\longleftrightarrow`	$\Longleftrightarrow$	`\Longleftrightarrow`
$\mapsto$	`\mapsto`	$\longmapsto$	`\longmapsto`
$\uparrow$	`\uparrow`	$\downarrow$	`\downarrow`
$\nearrow$	`\nearrow`	$\searrow$	`\searrow`
$\swarrow$	`\swarrow`	$\nwarrow$	`\nwarrow`
$\hookrightarrow$	`\hookrightarrow`	$\hookleftarrow$	`\hookleftarrow`
$\rightharpoonup$	`\rightharpoonup`	$\leftharpoonup$	`\leftharpoonup`
$\rightleftharpoons$	`\rightleftharpoons`	$\leftrightharpoons$	`\leftrightharpoons`
$\leadsto$	`\leadsto`	$\rightsquigarrow$	`\rightsquigarrow`
$\xrightarrow{p}$	`\xrightarrow {p}`	$\xleftarrow{d}$	`\xleftarrow {d}`

Binary Operators

Symbol	LaTeX input	Symbol	LaTeX input
$\pm$	`\pm`	$\mp$	`\mp`
$\times$	`\times`	$\div$	`\div`
$\cdot$	`\cdot`	$*$	`\ast`
$\star$	`\star`	$\circ$	`\circ`
$\bullet$	`\bullet`	$\diamond$	`\diamond`
$\oplus$	`\oplus`	$\ominus$	`\ominus`
$\otimes$	`\otimes`	$\oslash$	`\oslash`
$\odot$	`\odot`	$\bigoplus$	`\bigoplus`
$\bigotimes$	`\bigotimes`	$\wedge$	`\wedge`
$\vee$	`\vee`	$\cap$	`\cap`

Symbol	LaTeX input	Symbol	LaTeX input
$\cup$	`\cup`	$\sqcap$	`\sqcap`
$\sqcup$	`\sqcup`	$\setminus$	`\setminus`

Delimiters and Brackets

Symbol	LaTeX input	Symbol	LaTeX input
(x)	`(x)`	$[x]$	`[x]`
$\{x\}$	`\{x\}`	$\langle x\rangle$	`\langle x\rangle`
$\lvert x\rvert$	`\lvert x\rvert`	$\lVert x\rVert$	`\lVert x\rVert`
$\lfloor x\rfloor$	`\lfloor x\rfloor`	$\lceil x\rceil$	`\lceil x\rceil`
$\left(\frac{a}{b}\right)$	`\left (\frac {a}{b}\right )`	$\left\{x:\ x>0\right\}$	`\left \{x:  x>0\right \}`
$\bigl(x\bigr)$	`\bigl ( x \bigr )`	$\Bigl(x\Bigr)$	`\Bigl ( x \Bigr )`
$\biggl(x\biggr)$	`\biggl ( x \biggr )`	$\left[x\right]$	`\left [ x \right ]`

Accents, Decorations, and Fonts

Symbol	LaTeX input	Symbol	LaTeX input
$\hat{x}$	`\hat {x}`	$\widehat{XYZ}$	`\widehat {XYZ}`
$\bar{x}$	`\bar {x}`	$\overline{XYZ}$	`\overline {XYZ}`
$\tilde{x}$	`\tilde {x}`	$\widetilde{XYZ}$	`\widetilde {XYZ}`
$\vec{x}$	`\vec {x}`	$\dot{x}$	`\dot {x}`
$\ddot{x}$	`\ddot {x}`	$\breve{x}$	`\breve {x}`
$\check{x}$	`\check {x}`	$\acute{x}$	`\acute {x}`
$\grave{x}$	`\grave {x}`	$\mathbb{R}$	`\mathbb {R}`
$\mathcal{F}$	`\mathcal {F}`	$\mathfrak{g}$	`\mathfrak {g}`
$\mathbf{x}$	`\mathbf {x}`	d	`\mathrm {d}`
X	`\mathsf {X}`	$\mathtt{x}$	`\mathtt {x}`
$\boldsymbol{\theta}$	`\bm {\theta }`	$\boldsymbol{\Omega}$	`\boldsymbol {\Omega }`

Miscellaneous Symbols

Symbol	LaTeX input	Symbol	LaTeX input
∞	`\infty`	∂	`\partial`
∇	`\nabla`	$\hbar$	`\hbar`
ℓ	`\ell`	$\wp$	`\wp`
$\Re$	`\Re`	$\Im$	`\Im`
$\angle$	`\angle`	$\measuredangle$	`\measuredangle`
$\triangle$	`\triangle`	$\square$	`\square`
$\lozenge$	`\lozenge`	$\Diamond$	`\Diamond`
$\star$	`\star`	$\bigstar$	`\bigstar`
$\top$	`\top`	$\bot$	`\bot`
$\vdots$	`\vdots`	$\ddots$	`\ddots`
$\cdots$	`\cdots`	$\ldots$	`\ldots`